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Code to Connect Hackathon 2026

AI/NLP Risk Engine: From Financial News to Portfolio Signals

End-to-End Tactical Index Rebalancing with Transparent Explainability

PARAMETER	SUBMISSION DETAILS
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Hackathon Event	Code to Connect Hackathon 2026 • Final Round Submission
Implemented Downstream	Module A (Tactical Index Rebalancer) • 14 Large-Cap Equities

EXECUTIVE PROBLEM STATEMENT & SCOPE

Systematic Conversion of Unstructured Text to Portfolio Risk

Bridging financial NLP directly into quantitative risk constraints and index weights

PROBLEM VS. ENGINEERING SOLUTION MATRIX

INDUSTRY CHALLENGE	AI RISK ENGINE SOLUTION
Information Overload	Multi-source ingestion pipeline processes **tweets, NewsAPI, and GDELT** into a unified schema with cross-label deduplication.
Black-Box LLM Distrust	Explainable linear impact score (1.0 to 10.0) built from **4 audit-ready named components** configurable via YAML.
Execution Drag & Noise	Tactical rebalancer governed by **± 0.20 deadbands, 2-day EMA smoothing, and 10% daily turnover cap** to preserve net returns.
Offline Reproducibility	Deterministic committed sample datasets (<5 MB) allowing full audit and replay with **zero API keys or live downloads**.

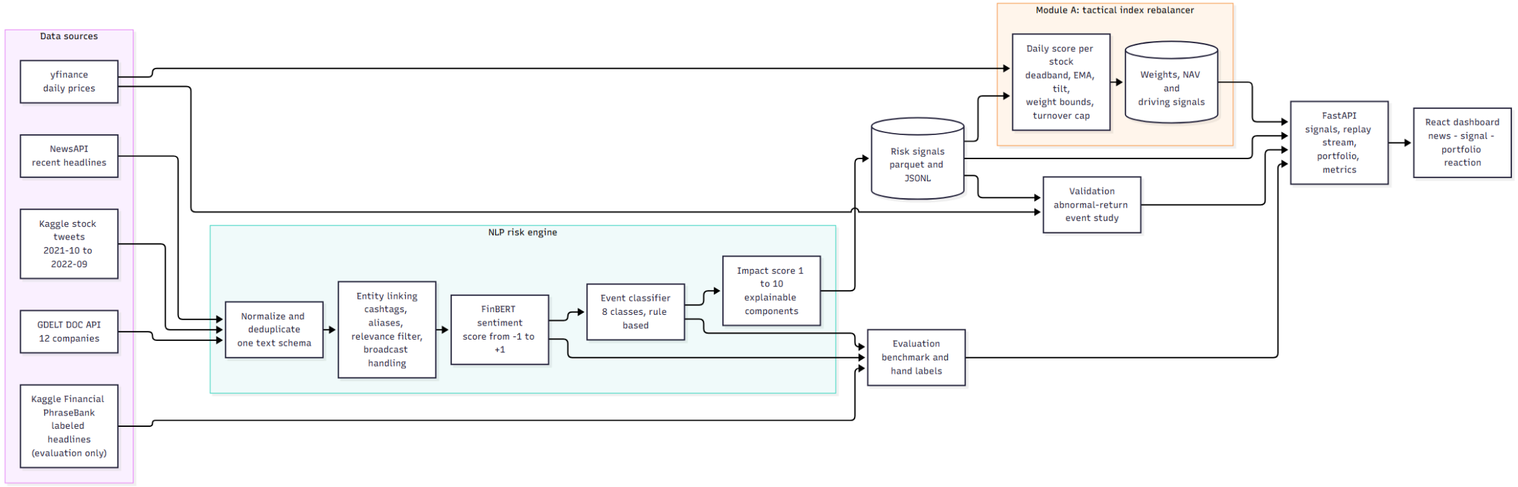
EMITTED SIGNAL SCHEMA SPECIFICATION

FIELD NAME	TYPE / RANGE	OPERATIONAL MEANING
`sentiment_score`	`[-1.0, +1.0]`	FinBERT signed score measuring tone (**negative** to **positive**).
`confidence`	`[0.0, 1.0]`	Softmax model conviction over target sentiment classification.
`event_type`	`8 Classes`	Rule-based: **Earnings**, **Credit Event**, **Regulatory**, Macro, M&A, etc.
`impact_score`	`[1.0, 10.0]`	Severity rank: Event (35%) + Sentiment (30%) + Credibility (20%) + Volume (15%).
`trading_action`	`Discrete`	Downstream suggestion: **BUY**, **REDUCE**, or **HOLD**.

END-TO-END SYSTEM PIPELINE

System Architecture: High-Resolution Data Pipeline

Modular architecture connecting raw financial text sources to downstream tactical index weights



LAYER	COMPONENT TECHNOLOGY	PRIMARY RESPONSIBILITY
1. Data Sources	Kaggle Stock Tweets (2021–22), NewsAPI, GDELT 2.0, yfinance	Ingest structured prices and raw text streams covering 14 core + 7 news-only equities.
2. NLP Risk Engine	Normalization, Cashtag Linking, FinBERT (CPU), Event Classifier	Deduplicate cross-label chatter, attribute symbols, score sentiment and 1–10 impact severity.
3. Module A Rebalancer	EMA Smoothing, Tilt k=1.5, Asymmetric Negative Multiplier (1.25x)	Map daily signals into bounded constituent weights [3.57%, 14.29%] with 10% turnover cap.
4. Delivery & Dashboard	FastAPI, Server-Sent Events (SSE), React 18, Apache ECharts	Interactive 3-column replay workspace (News Stream • Signal Inspector • Portfolio Reaction).

NLP RISK PIPELINE SPECIFICATION

Entity Linking, Sentiment Scoring & Impact Formula

Deterministic text normalization, domain-tuned FinBERT inference, and auditable scoring arithmetic

CORE NLP STAGES & OPERATIONS

STAGE	ALGORITHM / METHOD	OUTPUT / PERFORMANCE
Text Cleaning	Regex stripping of template spam, URL normalization, HTML entity decoding.	Single unified record schema across all sources.
Entity Linking	Cashtags (`$DIS`, `$AAPL`), aliases, company name dictionaries, broadcast splitting.	Clean ticker attribution; multi-ticker weight division.
FinBERT Sentiment	Pretrained `ProsusAI/finbert` with persistent SQLite disk cache.	Score $s \in [-1.0, +1.0]$, softmax confidence $c \in [0.0, 1.0]$.
Event Classification	Rule-based pattern matcher across 8 financial taxonomy classes.	Precision 1.0 on Credit/Regulatory/Earnings triggers.

IMPACT ARITHMETIC: LINEAR COMPONENT BREAKDOWN

COMPONENT	WEIGHT	SCALE FORMULA & CONFIG SOURCE
Event Severity	35%	Regulatory/Credit: `0.9`, Earnings: `0.7`, Product: `0.5`, Other: `0.3`
Sentiment Strength	30%	$\|s\| \times c$ (Absolute sentiment magnitude weighted by confidence)
Source Credibility	20%	GDELT: `0.85`, NewsAPI: `0.80`, Kaggle Tweets: `0.50`
Mention Volume	15%	Log-scaled ticker tweet density relative to rolling median
Final Severity Score	100%	**`Impact = 1.0 + 9.0 × ∑(w_i × Component_i) ∈ [1.0, 10.0]`**

Auditability Note: Every single impact score displayed on the dashboard can be unpacked via "Show the math" into its 4 raw component values.

TACTICAL INDEX REBALANCER LOGIC

Module A: Quantitative Rules, Parameters & Bounds

Dynamic conviction tilting across 14 large-cap US equities with institutional execution safeguards

REBALANCER MATHEMATICAL PARAMETERS (REBALANCER.YAML)

PARAMETER	VALUE	FINANCIAL RATIONALE & FORMULATION
w_base	7.14%	Equal-weight anchor across 14 stocks ($1 / 14 = 7.14\%$).
tilt_factor (k)	1.50	Conviction scalar: $w_{\text{raw}, i} = w_{\text{base}} \times (1.0 + k \times \text{EMA}_i)$.
ema_half_life	2.0 Days	Exponential memory filter preventing high-frequency noise churn.
deadband	±0.20	Threshold filter: scores inside $[-0.20, +0.20]$ zeroed out.
negative_multiplier	1.25×	Asymmetric risk aversion: negative headlines receive 25% higher penalty.

PORTFOLIO RISK BOUNDARIES & FRICTION SAFEGUARDS

CONSTRAINT	BOUNDARY	IMPLEMENTATION SAFEGUARD
Lower Weight Bound	3.57% (0.5×)	Floor prevents complete asset divestment or short exposures.
Upper Weight Bound	14.29% (2.0×)	Ceiling limits single-stock concentration risk (e.g. TSLA).
Turnover Cap	10% / Day	Clamps daily two-way volume to eliminate excessive transaction drag.
Transaction Frictions	5 bps	Institutional execution cost deducted on all turnover in net NAV.
Fallback Behavior	Equal Weight	Missing market data gracefully reverts allocations back to 7.14%.

ABNORMAL RETURN EVENT STUDY

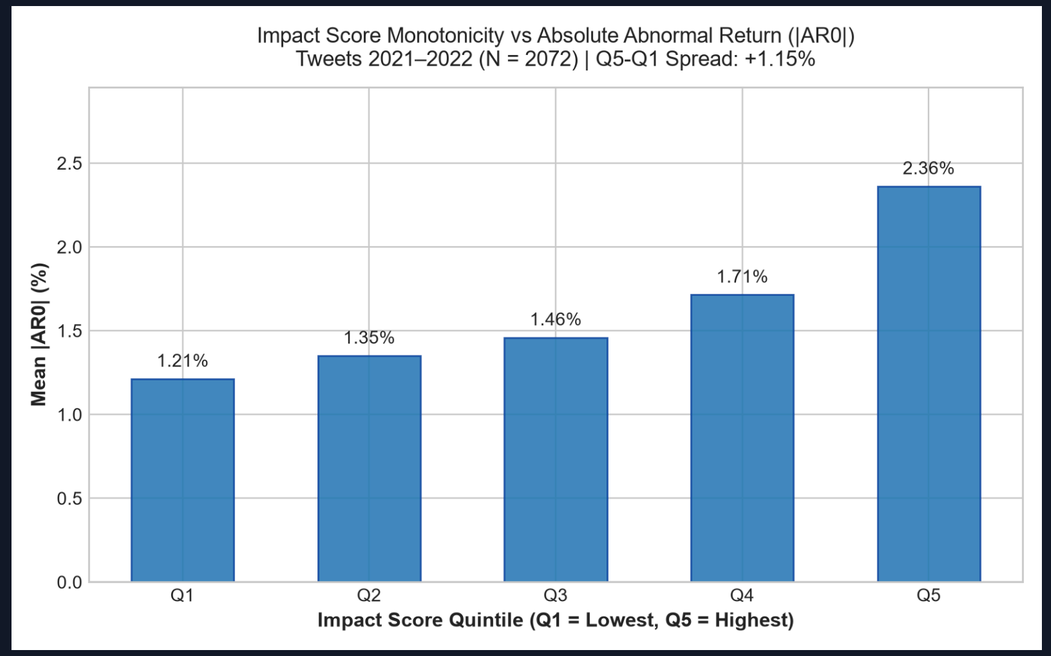
Empirical Validation: Impact Score vs. Real Price Moves

Event study across 2,072 ticker-day observations (2021–2022) with block bootstrap confidence intervals

IMPACT SCORE QUINTILES ON EVENT-DAY ABNORMAL MOVE (|AR0|)

QUINTILE	COUNT	MEAN IMPACT	MEAN |AR0|	MEAN |CAR(0,2)|
Q1 (Lowest)	415	2.70	1.21%	2.46%
Q2	414	3.50	1.35%	2.52%
Q3	415	4.63	1.46%	2.81%
Q4	414	5.15	1.71%	3.10%
Q5 (Highest)	414	6.36	**2.36%**	**3.89%**
Spread (Q5 - Q1)	2,072	**+3.66**	**+1.15%**	**+1.43%**

STATISTICAL METRIC	RESULT	INTERPRETATION
Spearman ρ (Impact vs |AR0|)	ρ = 0.159 (p = 3.08e-13)	Statistically significant monotonic correlation.
Volatility Control (Partial ρ)	ρ = 0.116 (p = 1.05e-07)	Survives control for recent 20-day historical vol.
Placebo Permutation (1,000x)	p = 0.0010	1,000 within-ticker shuffles confirm non-randomness.



MODULE A EMPIRICAL RESULTS

1-Year Historical Backtest & Honesty Diagnostics

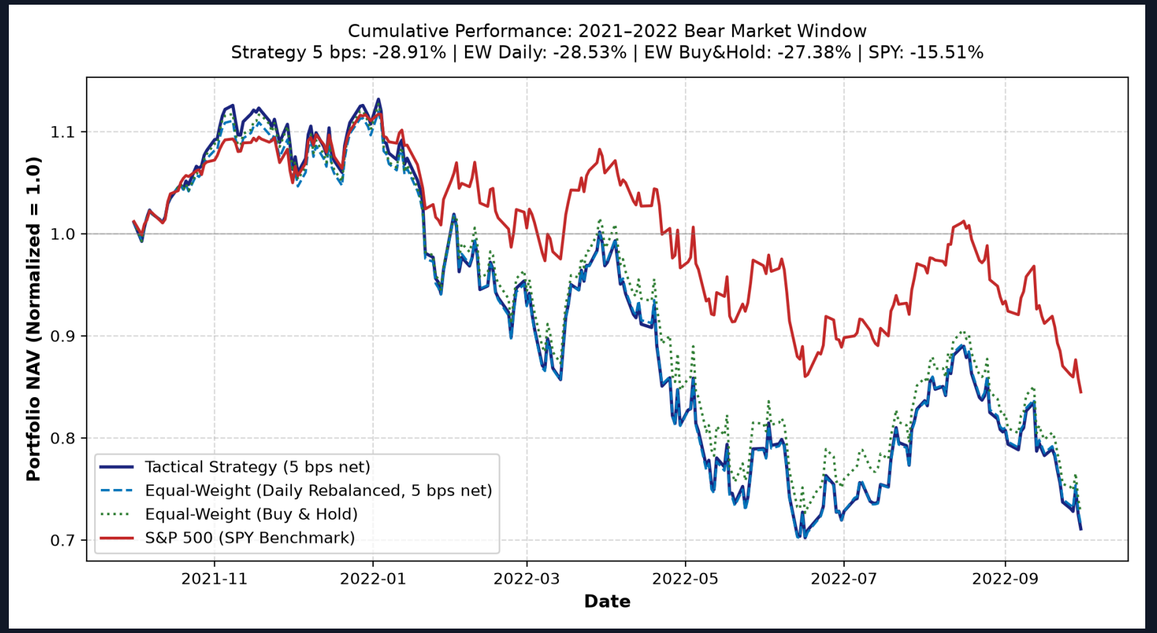
Performance over 252 trading days (Oct 2021 to Sep 2022) during the tech sector market downturn

STRATEGY COMPARISON MATRIX (5 BPS NET FRICTION)

STRATEGY	CUM. RETURN	SHARPE (RF=0)	MAX DRAWDOWN
Tactical Strategy (5 bps)	**-28.91%**	-0.9209	-37.95%
Equal Weight (Daily Rebal)	-28.53%	-0.9092	-37.39%
Equal Weight (Buy & Hold)	-27.38%	-0.9323	-35.56%
S&P 500 (SPY Benchmark)	-15.51%	-0.7085	-24.37%

HONESTY DIAGNOSTICS & PLACEBO PERCENTILES

DIAGNOSTIC TEST	FINDING	HONEST CONCLUSION
Placebo Permutation (200x)	**48.5th Percentile**	Strategy sits near center of random noise distribution.
TSLA Exclusion Test	-28.65% Net	Removing 70% tweet volume stock does not change result.
Sensitivity Grid (9 cells)	-28.56% to -29.20%	Flat returns across tilt $k \in [0.75, 3.0]$ and half-life $[1, 5]$d.



EVALUATION GAPS & FUTURE EXTENSIONS

Model Limitations, Learnings & Module B Readiness

Transparent appraisal of production boundaries and enterprise wholesale stress-testing extension

BENCHMARK VS. REAL-WORLD SOCIAL EVALUATION GAP

DATASET SOURCE	SAMPLE SIZE	FINBERT ACCURACY	MACRO F1
Public FinancialPhraseBank	4,838 Headlines	**88.9%**	**0.88**
Independent Hand Labels (All)	180 Texts	**61.1%**	**0.60**
• GDELT Curated Headlines	50 Articles	74.0%	0.72
• NewsAPI Articles	44 Articles	61.4%	0.61
• Twitter Social Chatter	86 Tweets	**53.5%**	**0.51**

Domain Learning: FinBERT exhibits high precision on bad news (**0.85**), making it reliable for risk detection, but misses subtle positive analyst targets (positive recall: **0.30**).

MODULE B: WHOLESALE LOAN STRESS TESTING ROADMAP

MODULE B PILLAR	RISK ENGINE INTEGRATION	ENTERPRISE RISK VALUE
Signal Feed Reuse	Zero engine modification: consumes the exact same `RiskSignal` feed.	Shared intelligence across both market equity desks and commercial credit risk teams.
Credit Migration (PD)	Severe **Credit Event** / **Regulatory** triggers bump Probability of Default.	Early warning rating downgrade detection ahead of public agency revisions.
Loss Given Default (LGD)	Macro event severities scale industry sector recovery rates down.	Dynamic collateral hair-cuts and wholesale portfolio provisions.
Expected Loss (EL)	$EL = EAD \times PD_{\text{adjusted}} \times LGD_{\text{adjusted}}$	Instant capital adequacy and regulatory stress reporting during sudden news shocks.